

PROGRAM C: OPTIMAL PAGE REPLACEMENT

C PROGRAM

```
#include <stdio.h>
```

```
int main() {  
    int pages[50], frames[10];  
  
    int n, f;  
  
    int i, j, k, pos;  
  
    int fault = 0;  
  
    int found;  
  
    printf("Enter Number of Pages: ");  
    scanf("%d", &n);  
  
    printf("Enter Reference String:\n");  
    for (i = 0; i < n; i++) {  
        scanf("%d", &pages[i]);  
    }  
  
    printf("Enter Number of Frames: ");  
    scanf("%d", &f);  
  
    for (i = 0; i < f; i++) {  
        frames[i] = -1;  
    }  
  
    for (i = 0; i < n; i++) {  
        found = 0;  
        for (j = 0; j < f; j++) {  
            if (frames[j] == pages[i]) {  
                found = 1;  
                break;  
            }  
        }  
    }  
}
```

```

    if (found == 0) {
        for (j = 0; j < f; j++) {
            int future = 999;
            for (k = i + 1; k < n; k++) {
                if (frames[j] == pages[k]) {
                    future = k;
                    break;
                }
            }
            if (j == 0 || future < pos) {
                pos = future;
            }
        }
        frames[0] = pages[i];
        fault++;
    }
}

printf("Total Page Faults = %d\n", fault);

return 0;
}

```

SHELL SCRIPT

```

GNU nano 8.7.1
#!/bin/bash

echo "Optimal Page Replacement Demonstration"

pages=(7 0 1 2 0 3 0 4 2 3 0 3 2)

echo "Reference String: ${pages[@]}"

echo "Optimal Algorithm Executed"

```

Output :

```
(ganesh@LAPTOP-R3RDH9NG)~$ nano optimal.sh  
(ganesh@LAPTOP-R3RDH9NG)~$ chmod +x optimal.sh  
(ganesh@LAPTOP-R3RDH9NG)~$ bash optimal.sh  
Optimal Page Replacement Demonstration  
Reference String: 7 0 1 2 0 3 0 4 2 3 0 3 2  
Optimal Algorithm Executed
```